

EE #3

Joseph
Sensault

1) $\Delta V = 0$

$P_1 = 5 \text{ bar}$

$V_f = .005 \text{ m}^3$

$V_H = .00158 \text{ m}^3/\text{kg}$

$V_g = 1.5 \text{ m}^3$

$V_g = .2503 \text{ m}^3/\text{kg}$

$m_f = \frac{.005}{.00158} = 3.165 \text{ kg}$

$m_g = \frac{1.5}{.2503} = 5.993 \text{ kg}$

$x = \frac{m_g}{m_g + m_f} = \frac{5.993}{9.158} = .654$

$U_1 = 1916.39 + .654(1321.62 - 1916.39)$

$U_1 = 932.59 \text{ kJ/kg}$

$V = V_f + x(V_g - V_f)$

$V = .00158 + .654(.2503 - .00158)$
 $-.00158 + .163$

$4.38 = \frac{1330.64 - 1328.04}{1330.64 - U_2}$

$U_2 = 1330.05 \text{ kJ/kg}$

$U_1 = U_2 = .1646 \text{ m}^3/\text{kg}$

interpolate:

$\frac{.1596 - .1615}{-.1596 - .1646} = 4.38 = \frac{8 - 7}{8 - P_2}$

$P_2 = 7.77 \text{ bar}$

$W = -m(U_2 - U_1)$

$W = (3.165 + 5.993)(1330.05 - 932.59)$

$W = 3639.9 \text{ kJ}$

4.38

$$R = .465 \text{ kJ/kgK}$$

$$18 \text{ kg} = m$$

$$P_1 = 100 \times 10^3 \text{ Pa}$$

$$T = -10^\circ\text{C} = 263 \text{ K}$$

$$P_2 = 550 \times 10^3 \text{ Pa}$$

$$PV = mRT$$

$$100 \times 10^3 (V) = 18(.465)(263)$$

$$V = .023 \frac{\text{kg kJ/K}}{\text{kg/K Pa}}$$

$$V = .023 \frac{\text{kJ}}{\text{Pa}} = \frac{\text{N}^2 \text{m} \times 1000}{\text{N}}$$

$$V = 23 \text{ m}^3$$

$$P_1 V_1^{1.1} = P_2 V_2^{1.1}$$

$$\left(\frac{V_2}{V_1}\right)^{1.1 \frac{1}{1.1}} = \frac{P_1^{1/1.1}}{P_2^{1/1.1}}$$

$$\frac{V_2}{V_1} = \left(\frac{100 \times 10^3}{550 \times 10^3}\right)^{1/1.1}$$

$$V_2 = .023 \left(\frac{100}{550}\right)^{1/1.1}$$

$$V_2 = 4.88 \text{ m}^3$$

$$W = \frac{P_2 V_2 - P_1 V_1}{1 - n}$$

$$W = \frac{(550 \times 10^3)(4.88) - (100 \times 10^3)(23.11)}{-0.1}$$

$$W = -3730000 \text{ J}$$

$$W = -3730 \text{ kJ}$$

$$Q = -2929.1 \text{ to find Internal Energy}$$

$$Q = -3730 + (18 \cdot 1.6 (307.1 - 263))$$

$$Q = -3730 + \Delta U$$

$$\frac{T_1}{T_2} = \left(\frac{P_1}{P_2}\right)^{\frac{1}{1.1}}$$

$$\Delta U = m C_v (T_2 - T_1)$$

$$T_2 = 307.1$$

$$R = .287 \frac{\text{KJ}}{\text{kg K}}$$

$$3) m = .25 \text{ kg}$$

$$p = 2 \text{ bar}$$

$$P = 2 \text{ bar}$$

$$T = 473 \text{ K}$$

$$V = 2V_1$$

$$PV = mRT$$

$$W = -P(V_2 - V_1)$$

$$W = -200000(.17)$$

$$W = 34000 \text{ J}$$

$$W = -34 \text{ KJ}$$

$$200000 V_1 = .25 (.287) (473)$$

$$V_1 = .00017 \frac{\text{kg KJ K}}{\text{kg K Pa}}$$

$$V_1 = .00017 \frac{\text{KJ}}{\text{Pa}} = \frac{\text{m}^2 \text{Nm} \times 1000}{\text{N}}$$

$$V_1 = .17 \text{ m}^3$$

$$V_2 = .34$$

$$C = 1.005 \frac{\text{KJ}}{\text{kg K}}$$

$$Q = mC(T_2 - T_1)$$

$$\frac{T_2}{T_1} = \frac{V_2}{V_1}$$

$$T_2 = 2(473)$$

$$T_2 = 946 \text{ K}$$

$$R = \frac{\bar{R}}{M} = \frac{8.314}{28.97}$$

$$PV = mRT$$

$$200000 V_1 = .25 (.287) (473)$$

$$V_1 = .00017 \frac{\text{kg KJ K}}{\text{kg K Pa}}$$

$$V_1 = .17 \text{ m}^3$$

$$Q = (.25)(1.005)(946 - 473)$$

$$Q = 118.8$$

$$PV^{1.25} = C$$

4)

$$P_1 = 7 \text{ bar}$$

$$T_1 = 275 \text{ K}$$

$$V_1 = 0.25 \text{ m}^3$$

$$V_2 = 0.03$$

$$W = \frac{P_2 V_2 - P_1 V_1}{-1.25}$$

$$W = \frac{99000(0.03) - (70000)(0.25)}{-1.25}$$

$$70000(0.25)^{1.25} = P_2(0.03)^{1.25}$$

$$P_2 = 9.9 \text{ bar}$$

$$W = -48800 \text{ J}$$

$$W = -48.8 \text{ kJ}$$

$$Q = W + \Delta U \quad \text{and} \quad \Delta U = m C_v (T_2 - T_1) \quad \text{and} \quad C_{v, \text{air}} = 0.794 \frac{\text{kJ}}{\text{kg K}}$$

$$\frac{V_2}{V_1} = \left(\frac{P_1}{P_2} \right)^{\frac{1}{n}}$$

$$\frac{T_1}{T_2} = \left(\frac{P_1}{P_2} \right)^{\frac{n-1}{n}}$$

$$\left(\frac{0.03}{0.25} \right) = \left(\frac{70000}{P_2} \right)^{\frac{1}{1.25}}$$

$$\frac{275}{T_2} = \left(\frac{70000}{99150} \right)^{\frac{0.25}{1.25}}$$

$$\left(\frac{0.03}{0.25} \right)^{1.25} = \frac{70000}{P_2}$$

$$T_2 = 467.3 \text{ K}$$

$$0.0706 P_2 = 70000$$

$$P_2 = 99150 \text{ Pa}$$

How to find m??

$$Q = -48.8 + m(0.794)(467.3 - 275)$$